

Langflow Agent Builder RAG Handbook

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Section 1. Why RAG has two lifecycles

RAG should be operated as two lifecycles, not as one giant flow.

The ingestion flow handles upload, parse, chunk, embedding, and Milvus upsert.

The user runtime flow handles Chat Input, query embedding, Milvus search, prompt, and answer.

This separation makes re-indexing, deployment, monitoring, and testing easier.

Expected question:

Why should RAG be split into an ingestion flow and a user runtime flow?

Expected answer:

The ingestion flow updates the knowledge base, while the runtime flow answers user questions.

Section 2. Component choices for beginners

Read File is best for PDF, DOCX, PPTX, and document extraction.

File Dataset Loader is best for CSV, JSON, JSONL, Markdown, and Excel datasets.

The loader normalizes datasets into row_count, columns, preview_rows, rows, and warnings.

Run Flow Payload Adapter serializes structured Data payloads into a JSON Message.

Use it when a subflow accepts text or Message input instead of a full Data object.

Route Gate should mark inactive branches as success=false, active=false, skipped=true.

Downstream nodes should process only active=true and skipped=false branches.

Section 3. Milvus ingest_data record contract

Each Milvus record should include id, text, page_content, and metadata.

The metadata should include document_id, source_file, file_type, page, chunk_index, content_type, modalities, parse_mode, and source_locator.

Recommended collection name for this sample: agent_builder_study_docs.

Recommended primary key field: id.

Recommended text fields: text and page_content.

Recommended metadata fields: document_id, source_file, page, chunk_index, source_locator.

Operational rule:

Do not push full large tables into prompts. Prefer row_count, columns, preview_rows, and data_ref.